

Solving $Ax = 0$: Pivot Variables, Special Solutions

Lesson 6 said what the do-nothing trades are. This lesson is a four-move recipe that finds every one of them, for any menu of funds, even when there are more funds than rules.

PART 1 Index

One question runs through the whole lesson: **which trades change nothing?** Each topic is one move of the recipe that answers it.

1. The question, and a guess

Four funds, three rules. We can guess two do-nothing trades. Are there more? The recipe will tell us for sure.

2. Move 1: simplify the rules

Elimination, explained as rewriting the rules ("return beyond 12%"). Why that can't change the do-nothing trades.

3. Move 2: pivot columns and free columns

Which amounts you choose and which the rules then force. Rank.

4. Move 3: the special solutions

Choose the free amounts one at a time. Out come exactly our two guesses.

5. Move 4: every do-nothing trade

Every do-nothing trade is a mix of the special solutions, and nothing else exists. With a try-it box.

6. The same four moves on smaller menus

A 2×2 and two 3×3 s. How many special solutions: $n - r$.

7. The shortcut: the reduced form R

Two more row operations, and the special solutions can be read straight off.

METHODS IN THIS LESSON

Row operations keep $N(A)$

Echelon form U

Pivot and free variables

Rank r

Special solutions

$n - r$ rule

Reduced row echelon form R

Nullspace matrix N

PART 2 Prerequisites

Click any link to review it, then come back.

- The nullspace (Lesson 6): the trades with $Ax = 0$, which change nothing.
- Elimination, 3×3 and back substitution (Lesson 2).
- Gauss–Jordan (Lesson 3): clearing above the pivots as well as below.

PART 3 Explanation

THE WHOLE LESSON IN ONE LINE

To find every trade that changes nothing: (1) simplify the rules by elimination, (2) see which fund amounts you can choose freely, (3) choose them one at a time to get a few basic do-nothing trades, (4) mix those to get all of them.

1. The question, and a guess

In Lesson 6 a menu of three funds had one do-nothing trade, and we found it by hand. Real menus are bigger: many products, a few things you measure. So we need a recipe that works for **any** menu. Here is the menu we'll use for the whole lesson.

THE MENU: FOUR FUNDS, THREE RULES

Fund	What ₹1 lakh does to (budget, return, exposure)	Note
Nifty	(1, 12, 1)	12% return, beta 1
Nifty ETF	(1, 12, 1)	a twin of Nifty from another fund house
Bonds	(1, 7, 0)	7% return, beta 0
Balanced	(1, 9.5, 0.5)	half Nifty, half bonds

A trade is four amounts $x = (x_1, x_2, x_3, x_4)$: ₹ lakh into Nifty, ETF, bonds, balanced. Positive = buy, negative = sell.

Each fund is a column, each rule is a row:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix} \begin{array}{l} \leftarrow \text{budget} \\ \leftarrow \text{return} \\ \leftarrow \text{exposure} \end{array}$$

The question: which trades x change nothing, so that $Ax = 0$? (That set is the nullspace $N(A)$.)

A guess

Just by looking at the menu, two do-nothing trades jump out:

1 GUESS 1

The ETF is a copy of Nifty. **Buy ₹1 lakh ETF, sell ₹1 lakh Nifty:**
 $(-1, 1, 0, 0)$.

CHECK

Budget $-1 + 1 = 0$. Return $-12 + 12 = 0$. Exposure $-1 + 1 = 0$.
 Nothing changes ✓.

2 GUESS 2

Balanced is half Nifty, half bonds. **Buy ₹1 lakh balanced, sell ₹0.5 lakh Nifty and ₹0.5 lakh bonds:** $(-0.5, 0, -0.5, 1)$.

CHECK

Budget $-0.5 - 0.5 + 1 = 0$. Return $-6 - 3.5 + 9.5 = 0$. Exposure $-0.5 + 0 + 0.5 = 0$ ✓.

But guessing leaves two questions open. **Are there other do-nothing trades we missed?** And what do we do with a menu where nothing is obvious? The recipe answers both. Watch for these two guesses: the recipe will produce them on its own.

2. Move 1: simplify the rules

Three rules about four funds are hard to read. Elimination rewrites them in a simpler form. First, let's see why that's allowed.

Each row is a question about the trade

Row	The question it asks	Its formula
1	How much does the budget change?	$x_1 + x_2 + x_3 + x_4$
2	How much does the return change?	$12x_1 + 12x_2 + 7x_3 + 9.5x_4$
3	How much does the exposure change?	$x_1 + x_2 + 0x_3 + 0.5x_4$

A do-nothing trade is one where **every answer is 0**.

Why rewriting the questions is safe

Elimination replaces a question by a combination of questions, like "return minus $12 \times$ budget". If a trade answers 0 to both "return" and "budget", it answers $0 - 12 \times 0 = 0$ to the new question. And the step can be undone (return = new question + $12 \times$ budget), so no new trade can pass that didn't pass before.

RULE · ROW OPERATIONS KEEP THE NULLSPACE

Subtracting a multiple of one row from another, swapping rows, or dividing a row by a non-zero number never changes which trades give all zeros. So $Ax = 0$ and the simplified $Ux = 0$ have exactly the same solutions. (They do change the *other* answers, so never read reachable targets off U : that's Lesson 8's job.)

Doing it: three row operations

1 WHAT

Row 2 $- 12 \times$ row 1.

ENTRIES

$12 - 12(1) = 0$, $12 - 12(1) = 0$, $7 - 12(1) = -5$, $9.5 - 12(1) = -2.5$. New row 2: $(0, 0, -5, -2.5)$.

MEANING

The new question is "how much return **beyond** what the same money would earn in Nifty at 12%?" Nifty and its ETF: 0 extra. Bonds: 5 less. Bonded: 2.5 less.

2 WHAT

Row 3 $- 1 \times$ row 1.

ENTRIES

$1 - 1 = 0$, $1 - 1 = 0$, $0 - 1 = -1$, $0.5 - 1 = -0.5$. New row 3: $(0, 0, -1, -0.5)$.

MEANING

"Exposure minus budget": how much less exposure than an all-Nifty trade of the same size.

3 WHAT

Row 3 $- 0.2 \times$ row 2 (multiplier = $(-1)/(-5) = 0.2$).

ENTRIES

$0 - 0.2(0) = 0$, $0 - 0.2(0) = 0$, $-1 - 0.2(-5) = -1 + 1 = 0$, $-0.5 - 0.2(-2.5) = -0.5 + 0.5 = 0$. New row 3: $(0, 0, 0, 0)$.

MEANING

Row 3 now asks a question whose answer is **always** 0. For these funds, exposure carries no information that budget and return don't already give.

METHOD · ECHELON FORM U (THE STAIRCASE)

Eliminate column by column, as in Lesson 2. The result U is a staircase: each row starts further right than the row above, and rows that became all zeros sit at the bottom.

$$U = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & -5 & -2.5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{row 1: } x_1 + x_2 + x_3 + x_4 = 0 \\ \text{row 2: } -5x_3 - 2.5x_4 = 0 \\ \text{row 3: } 0 = 0 \end{array}$$

Same do-nothing trades as A , but now only **two** real rules, and the second one doesn't involve Nifty or the ETF at all. That's what makes the next move easy.

IN ONE LINE

Elimination rewrites the rules in a simpler form without changing which trades pass them.

3. Move 2: pivot columns and free columns

Now read the staircase U . In each non-zero row, look at its **first** non-zero entry. That entry is a **pivot**.

$$U = \begin{bmatrix} \boxed{1} & 1 & 1 & 1 \\ 0 & 0 & \boxed{-5} & -2.5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{row 1 starts in column 1 (Nifty)} \\ \text{row 2 starts in column 3 (bonds)} \\ \text{row 3 has no pivot} \end{array}$$

1 PIVOT COLUMNS

Columns 1 and 3: **Nifty and bonds**.

FREE COLUMNS

Columns 2 and 4, where no row starts: **ETF and balanced**.

2 SOLVE ROW 2

$-5x_3 - 2.5x_4 = 0 \Rightarrow x_3 = -0.5x_4$. The bond amount is decided by the balanced amount.

SOLVE ROW 1

$x_1 = -x_2 - x_3 - x_4 = -x_2 + 0.5x_4 - x_4 = -x_2 - 0.5x_4$. The Nifty amount is decided by the ETF and balanced amounts.

RULE · PIVOT VARIABLES AND FREE VARIABLES

The amounts in **free columns** (ETF x_2 , balanced x_4) can be anything you like: they are **free variables**. The amounts in **pivot columns** (Nifty x_1 , bonds x_3) are then forced by back substitution: they are **pivot variables**.

In plain words: whatever you do with the ETF and the balanced fund, the rules tell you exactly how much Nifty and bonds to trade to undo its effect. Buy ₹1 lakh ETF? You must sell ₹1 lakh Nifty. Buy ₹1 lakh balanced? You must sell ₹0.5 lakh bonds and ₹0.5 lakh Nifty.

Why these two are the free ones. Reading the menu left to right, a column is free when it adds **nothing new** compared with the columns before it. The ETF copies Nifty; balanced is a mix of Nifty and bonds. Nifty and bonds each brought something new, so they got the pivots. These are exactly the two overlaps we guessed.

DEFINITION · RANK R = THE NUMBER OF PIVOTS

Here $r = 2$. The rank counts the funds that bring something new, and equally the rules that really bite: of three rules, only two carried information.

IN ONE LINE

Pivot columns = amounts the rules force; free columns = amounts you choose. Rank = number of pivots.

4. Move 3: the special solutions

Move 2 says: choose the ETF and balanced amounts, and the rest follows. The simplest choices are to switch on **one free fund at a time**.

METHOD · SPECIAL SOLUTIONS

For each free variable in turn: set it to 1, set every other free variable to 0, and work out the pivot variables. Each free column gives one special solution.

1 CHOOSE

ETF $x_2 = 1$, balanced $x_4 = 0$.

BONDS

$$x_3 = -0.5x_4 = -0.5(0) = 0.$$

NIFTY

$$x_1 = -x_2 - 0.5x_4 = -1 - 0 = -1.$$

RESULT

$s_1 = (-1, 1, 0, 0)$. That's **guess 1**: swap Nifty for its twin.

2 CHOOSE

ETF $x_2 = 0$, balanced $x_4 = 1$.

BONDS

$$x_3 = -0.5(1) = -0.5.$$

NIFTY

$$x_1 = -0 - 0.5(1) = -0.5.$$

RESULT

$s_2 = (-0.5, 0, -0.5, 1)$. That's **guess 2**: build the balanced fund yourself.

Full check in the original rules, entry by entry:

	Budget 1, 1, 1, 1	Return 12, 12, 7, 9.5	Exposure 1, 1, 0, 0.5
$s_1 = (-1, 1, 0, 0)$	$-1 + 1 + 0 + 0 = 0 \checkmark$	$-12 + 12 + 0 + 0 = 0 \checkmark$	$-1 + 1 + 0 + 0 = 0 \checkmark$
$s_2 = (-0.5, 0, -0.5, 1)$	$-0.5 + 0 - 0.5 + 1 = 0 \checkmark$	$-6 + 0 - 3.5 + 9.5 = 0 \checkmark$	$-0.5 + 0 + 0 + 0.5 = 0 \checkmark$

The recipe found our two guesses without any guessing. On a menu where nothing is obvious, it works exactly the same way.

IN ONE LINE

One free amount = 1, the others = 0, solve for the rest: one special solution per free column.

5. Move 4: every do-nothing trade

Now the open question from topic 1: **are there any others?**

1 ANY CHOICE

Choose any ETF amount c and any balanced amount d . Move 2 forces $x_1 = -c - 0.5d$ and $x_3 = -0.5d$.

SPLIT IT

$$\begin{bmatrix} -c - 0.5d \\ c \\ -0.5d \\ d \end{bmatrix} = c \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} -0.5 \\ 0 \\ -0.5 \\ 1 \end{bmatrix} = c s_1 + d s_2.$$

2 SO

Every do-nothing trade is $c s_1 + d s_2$: "do guess 1, c times, and guess 2, d times".

NOTHING ELSE

A do-nothing trade is pinned down by its ETF and balanced amounts. If both are 0, the forced amounts are 0 too, so the trade is zero. There's no room for a hidden third kind.

3 EXAMPLE

Someone hands you the trade $(-3, 2, -1, 2)$. Its ETF amount is 2, its balanced amount is 2, so it must be $2s_1 + 2s_2 = (-2 - 1, 2, -1, 2) = (-3, 2, -1, 2) \checkmark$.

CHECK

Budget $-3 + 2 - 1 + 2 = 0$. Return $-36 + 24 - 7 + 19 = 0$.
Exposure $-3 + 2 + 0 + 1 = 0 \checkmark$.

Two free choices give two independent directions, so $N(A)$ is a **plane** inside $\mathbb{R}^4$ (the space of four-fund trades).

Try it: choose the free amounts

ETF amount $x_2 = c$

Balanced amount $x_4 = d$

Free (you chose): ETF $x_2 = 2$, balanced $x_4 = 2$.

Forced: Nifty $x_1 = -c - 0.5d = -3$, bonds $x_3 = -0.5d = -1$.

Trade $= 2s_1 + 2s_2 = (-3, 2, -1, 2)$ (₹ lakh in Nifty, ETF, bonds, balanced).

Budget: $1(-3) + 1(2) + 1(-1) + 1(2) = 0 \checkmark$

Return: $12(-3) + 12(2) + 7(-1) + 9.5(2) = 0 \checkmark$

Exposure: $1(-3) + 1(2) + 0(-1) + 0.5(2) = 0 \checkmark$

Nothing changes: every rule stays at 0, whatever free amounts you pick.

IN ONE LINE

Every do-nothing trade is a mix of the special solutions, and every mix is one: $N(A) = \text{all } cs_1 + ds_2$.

6. The same four moves on smaller menus

The recipe doesn't depend on the menu. Here it is on a 2×2 and two 3×3 s, move by move.

2×2 : Nifty and its twin, budget and return only

1 MOVE 1

$\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix}$. Row 2 $- 12 \times$ row 1: $12 - 12 = 0, 12 - 12 = 0$. $U = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$.

MOVE 2

Pivot in column 1 (Nifty); column 2 (ETF) is free. Rank 1.

MOVE 3

ETF = 1: row 1 gives $x_1 + 1 = 0$, so $x_1 = -1$. Special solution $(-1, 1)$.

MOVE 4

All do-nothing trades: $c(-1, 1)$, a line.

3 × 3: Nifty, bonds, balanced (Lesson 6's menu)

1 MOVE 1

Same three row operations as topic 2 (just without the ETF column):

$$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & -2.5 \\ 0 & 0 & 0 \end{bmatrix}.$$

MOVE 2

Pivots in columns 1 and 2 (Nifty, bonds); column 3 (balanced) is free. Rank 2.

MOVE 3

Balanced = 1. Row 2: $-5x_2 - 2.5(1) = 0 \Rightarrow x_2 = -0.5$. Row 1: $x_1 - 0.5 + 1 = 0 \Rightarrow x_1 = -0.5$. Special solution $(-0.5, -0.5, 1)$.

MOVE 4

All: $c(-0.5, -0.5, 1)$, a line. ($c = -2$ gives Lesson 6's $(1, 1, -2)$.)

3 × 3: Nifty, bonds, midcap

1 MOVE 1

Row 2 $- 12 \times$ row 1: $(0, -5, 15 - 12 = 3)$. Row 3 $-$ row 1: $(0, -1, 1.2 - 1 = 0.2)$. Row 3 $- 0.2 \times$ row 2: $(0, 0, 0.2 - 0.6 = -0.4)$. $U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 0 & 0 & -0.4 \end{bmatrix}$.

MOVE 2

Pivots in all three columns. **No free column.** Rank 3.

MOVES 3–4

Nothing to choose, so the only do-nothing trade is $(0, 0, 0)$. Midcap brings something new; no fund repeats another.

How many special solutions?

Every column is either a pivot column or a free column, and each free column gives one special solution. So:

RULE · $N - R$ SPECIAL SOLUTIONS

With n funds (columns) and rank r : number of free columns $= n - r =$ number of special solutions. Since there can't be more pivots than rows, $r \leq m$. So if there are **more funds than rules** ($n > m$), at least one column is free and a do-nothing trade **always** exists.

Menu	m rules	n funds	rank r	$n - r$	Do-nothing trades
Nifty, twin (2×2)	2	2	1	1	a line
Nifty, bonds, balanced	3	3	2	1	a line
Nifty, bonds, midcap	3	3	3	0	only 0
Nifty, ETF, bonds, balanced	3	4	2	2	a plane

WHY A DESK CARES

Three things you measure and four or more products: there is **guaranteed** to be a way to switch between products that changes nothing you measure. That's where a desk looks for the cheaper route, and also where risks you *don't* measure can slip in unnoticed.

▼ Check yourself (MIT recitation): all solutions of $x - 5y + 2z = 0$.

Move 1: one row, already a staircase. Move 2: pivot in column 1 (x); y and z are free; rank 1, so $3 - 1 = 2$ special solutions. Move 3: $y = 1, z = 0$ gives $x = 5$, so $(5, 1, 0)$; $y = 0, z = 1$ gives $x = -2$, so $(-2, 0, 1)$. Move 4: all $c(5, 1, 0) + d(-2, 0, 1)$, a plane.

IN ONE LINE

Same four moves every time; the number of special solutions is funds minus pivots, $n - r$.

7. The shortcut: the reduced form R

In Move 2 we still had to solve row 2, then put that into row 1. If we push elimination two steps further, that solving is already done for us.

METHOD · REDUCED ROW ECHELON FORM R

Start from U . Divide each pivot row by its pivot (so every pivot becomes 1), then clear the entries **above** each pivot too (Gauss-Jordan, Lesson 3). The result is $R = \text{rref}(A)$: same do-nothing trades, simplest possible rules.

1 PIVOT 2 $\rightarrow$ 1

Row 2 $\div (-5)$: $0, 0, -5/(-5) = 1, -2.5/(-5) = 0.5$. Row 2 = $(0, 0, 1, 0.5)$.

CLEAR ABOVE

Row 1 has a 1 above that pivot. Row 1 $- 1 \times$ row 2: $1 - 0 = 1$, $1 - 0 = 1$, $1 - 1 = 0$, $1 - 0.5 = 0.5$. Row 1 = $(1, 1, 0, 0.5)$.

$$R = \begin{bmatrix} \boxed{1} & 1 & 0 & 0.5 \\ 0 & 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{row 1: } x_1 + x_2 + 0.5x_4 = 0 \Rightarrow x_1 = -x_2 - 0.5x_4 \\ \text{row 2: } x_3 + 0.5x_4 = 0 \Rightarrow x_3 = -0.5x_4 \end{array}$$

Each row now contains **one** pivot amount and only free amounts: it is already the Move 2 answer. So the special solutions can be read off directly:

1 READING RULE

Set one free amount to 1. Each pivot amount is **minus** the number in that free column of R , in the pivot's row.

ETF = 1

ETF column of R is $(1, 0)$ in rows 1, 2. So Nifty = -1 , bonds = 0: $s_1 = (-1, 1, 0, 0) \checkmark$.

BALANCED = 1

Balanced column of R is $(0.5, 0.5)$. So Nifty = -0.5 , bonds = -0.5 : $s_2 = (-0.5, 0, -0.5, 1) \checkmark$.

METHOD · NULLSPACE MATRIX N

Put the special solutions side by side as columns. N holds every do-nothing trade at once ($N \begin{bmatrix} c \\ d \end{bmatrix} = c s_1 + d s_2$), and $AN = 0$, $RN = 0$.

$$N = \begin{bmatrix} -1 & -0.5 \\ 1 & 0 \\ 0 & -0.5 \\ 0 & 1 \end{bmatrix} \quad \begin{array}{l} \leftarrow \text{Nifty} \\ \leftarrow \text{ETF} \\ \leftarrow \text{bonds} \\ \leftarrow \text{balanced} \end{array}$$

Check $RN = 0$, each row of R against each column of N :

	$s_1 = (-1, 1, 0, 0)$	$s_2 = (-0.5, 0, -0.5, 1)$
Row 1 of R : $(1, 1, 0, 0.5)$	$-1 + 1 + 0 + 0 = 0 \checkmark$	$-0.5 + 0 + 0 + 0.5 = 0 \checkmark$
Row 2 of R : $(0, 0, 1, 0.5)$	$0 + 0 + 0 + 0 = 0 \checkmark$	$0 + 0 - 0.5 + 0.5 = 0 \checkmark$
Row 3 of R : $(0, 0, 0, 0)$	$0 \checkmark$	$0 \checkmark$

▼ For the curious: Strang's compact form $R = [I \ F]$ and $N = [-F \ ; \ I]$

List the pivot columns first (Nifty, bonds) and the free columns after (ETF, balanced). Then the pivot part of R is the identity $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and the free part is $F = \begin{bmatrix} 1 & 0.5 \\ 0 & 0.5 \end{bmatrix}$. The reading rule above

says the pivot amounts are $-F$ and the free amounts are I , so in that order $N = \begin{bmatrix} -F \\ I \end{bmatrix}$, and $RN = I(-F) + F I = 0$.

MIT's lecture matrix $\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$ has the same shape as our menu: $R = \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, pivots in columns 1 and 3, special solutions $(-2, 1, 0, 0)$ and $(2, 0, -2, 1)$.

▼ **Check yourself (MIT problem 7.1):** find R , the rank and the special solutions of $\begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 4 & 1 & 7 \\ 2 & -2 & 11 & -3 \end{bmatrix}$.

Move 1: row 3 $- 2 \times$ row 1 $= (0, -12, -3, -21)$; row 3 $+ 3 \times$ row 2 $= (0, 0, 0, 0)$. Shortcut: row 2 $\div 4 = (0, 1, \frac{1}{4}, \frac{7}{4})$; row 1 $- 5 \times$ row 2 $= (1, 0, \frac{23}{4}, \frac{1}{4})$. So $R = \begin{bmatrix} 1 & 0 & 23/4 & 1/4 \\ 0 & 1 & 1/4 & 7/4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, rank 2, free columns 3 and 4. Reading rule: $(-\frac{23}{4}, -\frac{1}{4}, 1, 0)$ and $(-\frac{1}{4}, -\frac{7}{4}, 0, 1)$. Try it in the lab on the Examples tab.

IN ONE LINE

R is U finished off: each row says "pivot amount $= -(\text{free amounts} \times \text{numbers})$ ", so the special solutions are read straight off it.